The HILOTHERAPY® (controlled hand-foot cooling) Avoids chemotherapy-induced polyneuropathy (CIPN) and nail toxicity a procedure is becoming established in the everyday clinical routine of oncology

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Background

Chemotherapy-induced polyneuropathy (CIPN) is a common side effect of antineoplastic systemic therapies. Grade-specific Hilotherapy® (1st generation) prevents (prophylactically used) CIPN and nail toxicity under oncological therapies (Oneda et al 2020 & 2021, Coolbrandt et al 2022, Schaper et al 2019). It is already established as a supportive therapy in many oncology practices/clinics (Oneda et al 2020&2021, Coolbrandt et al 2022, Schaper et al 2022). A data analysis of 151 breast cancer patients showed that with prophylactic Hilotherapy®(1st generation), CIPN ≥ grade 2 could be avoided in 93.4% of patients receiving taxane-containing chemotherapy (Schaper et al.).

4 weeks after last chemotherapy (EOT) 95.4 % show no CIPN > grade 1.

The development of the 2nd generation of the cooling pressure cuff now makes it possible to lower the skin surface temperature by a further 10°C **(Fig. 3).** This raises the question of whether the effectiveness of Hilotherapy® can be increased again and whether the post-cooling time of 30 minutes can be saved, which could be a practical relief for everyday clinical practice.

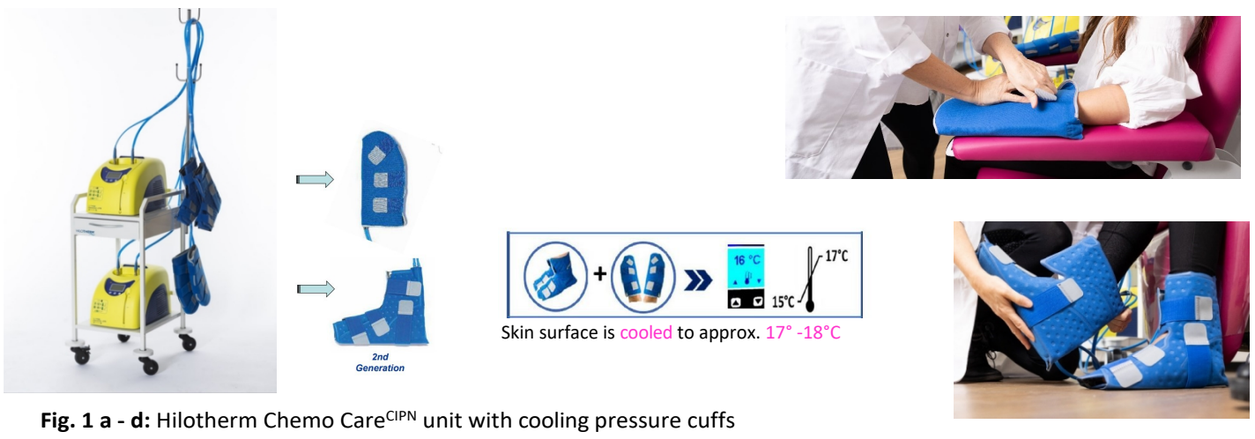


Fig. 1 a - d: Hilotherm Chemo Care[CIPN] unit with cooling pressure cuffs

Patients and method

The prophylactic use of Hilotherapy® 2nd generation (application with new pressure cooling cuffs) during taxane-containing chemotherapy for the treatment of breast cancer patients is being investigated. The object of the study is to optimise the effectiveness for CIPN prophylaxis and nail toxicity.

108 patients prophylactically used controlled hand-foot cooling (Hilotherapy® 2nd generation) to accompany their chemotherapy containing taxane. Hilotherapy® is a processor-controlled cooling method that is accurate to the degree. The Hilotherm ChemoCare device is equipped with hand/foot pressure cuffs and can be used with a cooling temperature accurate to the degree for constant cooling of the extremities. The device is operated with demineralised water and electricity (**Fig. 1a-d**).

As a standard procedure, hands and feet were cooled for 30 minutes before, during and 30 minutes after the end of chemotherapy at a device temperature setting of 15°-17°C. After each treatment, the symptoms of CIPN and nail toxicity were recorded with a questionnaire according to the CTCAE criteria.

Research approaches were:

- Survey of CIPN and nail toxicity
- Effectiveness 2nd generation (pressure cuff) of cooling cuffs
- Comparison of two experimental groups with and without post-cooling time

Group 1: 30 minutes pre-cooling time - cooling during chemotherapy - 30 minutes post-cooling time

Group 2: 30 minutes pre-cooling time - cooling during chemotherapy - no post-cooling time

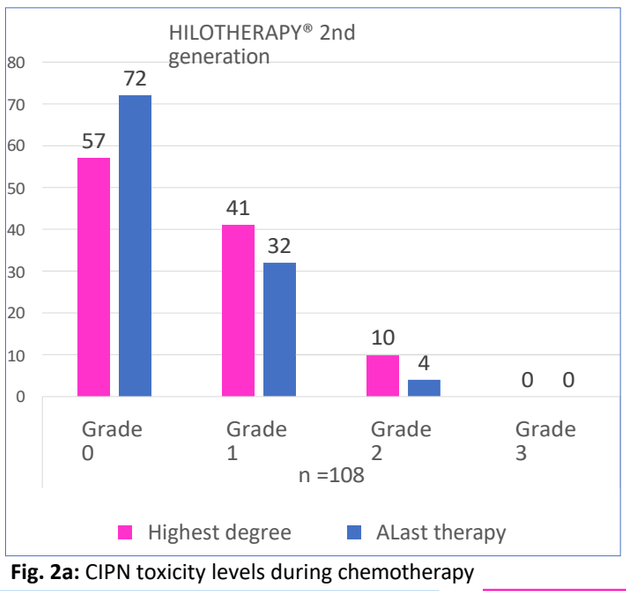


Fig. 2a: CIPN toxicity levels during chemotherapy

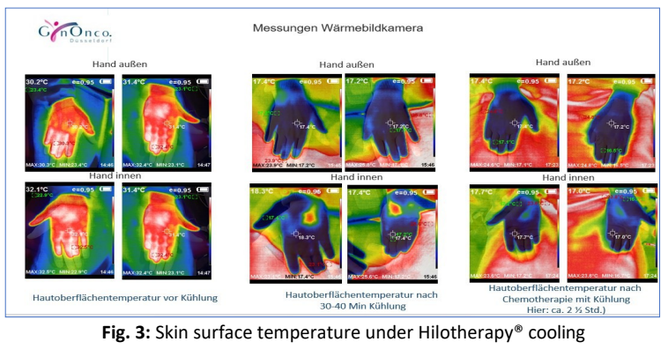


Fig. 3: Skin surface temperature under Hilotherapy® cooling

	2nd generation Hilotherapy® 15-17°C (n = 108)			
	Grade 0	Grade 1	Grade 2	Grade 3
hGAE	52,8%	38,0%	9,2%	0
Last therapy	66,7%	29,6%	3,7%	0

Fig. 2b: CIPN toxicity levels during chemotherapy in %.

Results: Effectiveness of Hilotherapy® 2nd generation

CIPN: Of 108 patients, 98 patients (90.8%) developed no or only mild symptoms of CIPN (grade 0: n=57; grade 1: n=41) during Hilotherapy® prophylaxis. 10 patients showed intermittent grade 2 toxicity, no patient developed grade 2 toxicity.

3 Symptoms. On the day of the last chemotherapy 104 patients (96.3%) showed no or mild symptoms (grade 0: n=72; grade 1: n=32), and 4 patients showed grade 2 toxicity. Grade 3 symptoms were not documented at any time point **(Fig. 2a&b). Nail toxicity:** At none of the 108 patients was nail toxicity was documented **(Fig. 2c).**

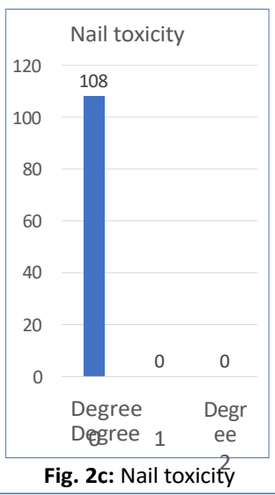


Fig. 2c: Nail toxicity

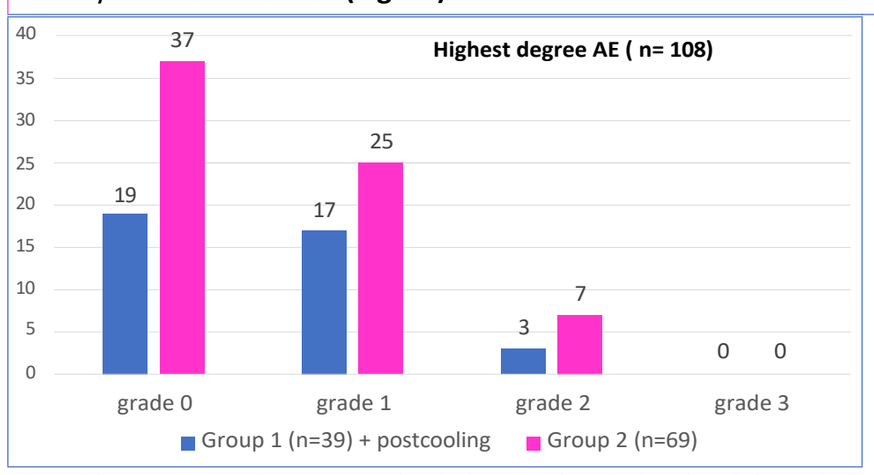
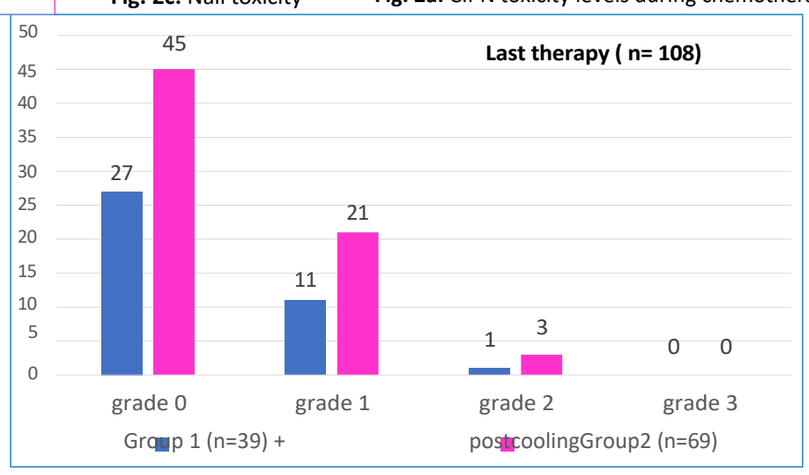


Fig. 4a&b: Comparison of CIPN toxicity levels during and on the last day of chemotherapy.



Results: Comparison of two experimental groups

Group 1: 30 min pre-cooling - cooling during chemotherapy - 30 min post-cooling

Of 39 patients in group 1, 36 patients developed no or only mild symptoms of CIPN during their chemotherapy treatment (grade 0: n=19, grade 1: n=17). 3 patients developed grade 2 toxicity and no patients developed grade 3 **(Fig. 4 a&b).** On the day of the last chemotherapy 38 patients showed no or mild CIPN symptoms (grade 0: n=27, grade 1: n=11) and 1 patient still grade 2.

Group 2: 30 min pre-cooling - cooling during chemotherapy - **no** post-cooling

Of 69 patients, 62 patients developed no or mild CIPN symptoms during treatment (grade 0: n=37, grade 1: n=25). 7 patients were documented with grade 2 toxicity, no patient with grade 3. On the day of the last chemotherapy session, 66 patients had no or mild symptoms (grade 0: n=45, grade 1: n=21) and 3 patients had grade 2 toxicity. The percentage comparison of both groups shows a slight numerical difference: On the day of the last chemotherapy, 97.4% of the patients in group 1 (with post-cooling) are without grade 2/3 toxicity. In group 2 (without post-cooling), 95.7% of the patients remain without grade 2/3 toxicity **(Fig. 5).**

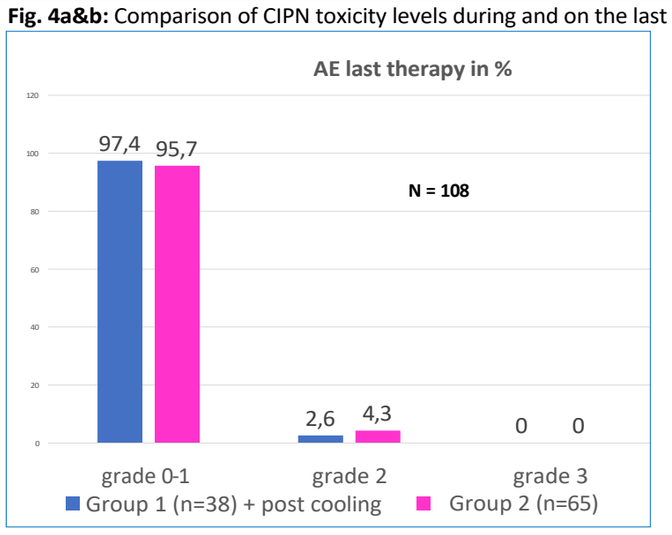


Fig. 5: Comparison of CIPN toxicity levels on the last day of chemotherapy

	Gloves / booties **1st generation** 10-12°C HILOTHERAPY last therapy	Gloves / booties **2nd generation** 15-17°C HILOTHERAPY Last therapy
Grade 0	52,3%	**66,7%**
Grade 1	40,4%	**29,6%**
Grade 2	6,6%	**3,7%**
Grade 3	0,6%	**0%**

Fig. 6: Comparison of CIPN toxicity levels Hilotherapy 1st generation vs. Hilotherapy® 2nd generation

Summary and discussion

With the prophylactic use of Hilotherapy® 1st & 2nd generation, CIPN toxicity > grade 1 can be avoided by 93% - 96% during chemotherapy **(Fig. 2a&b).** The development of the 2nd generation cooling pressure cuffs has led to a further slight optimisation of effectiveness **(Fig. 6).** Grade 3 toxicity no longer occurs. In the comparison of the two experimental groups with and without adherence to the post-cooling period of 30 minutes, minor numerical differences were found **(Fig. 6).** In group 1 (with post-cooling time), 97.4% of patients avoided CIPN > grade 1, in group 2 (without post-cooling time) 95.7% of patients remained without limiting CIPN symptoms (> grade 1). The new standard at GynOnco Düsseldorf integrates these results - post-cooling is dispensed with for the time being. This makes everyday practice in oncology easier. If a patient develops first symptoms of CIPN, the post-cooling period should be resumed.

The incidence rate of CIPN depends on several factors: Type of chemotherapy, dose, BMI, lack of physical activity and concomitant diseases. Seidmann et al (2018) reported grade 2&3 CIPN toxicity in 50% of patients receiving paclitaxel (80mg/m^2) weekly. CIPN can persist for many years (Blackley et al. 2019) and have a significant negative impact on the quality of life of oncology patients.

Modern oncological systemic therapies (individualised therapies, immunotherapies, targeted therapies...) lead to an improvement in the long-term prognosis of cancer patients - thus the importance of maintaining the quality of life increases. Controlled hand-foot cooling (Hilotherapy®) is a simple and very tolerable supportive therapy for patients to avoid CIPN. It is easy to implement for the nursing staff and does not significantly influence the daily clinical routine.

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Literature:

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Conflicts of Interest:

A. Kostara: has nothing to declare due to this investigation.

T. Schaper: Lecture fees, advisory boards, consulting contracts Hilotherm GmbH